



提取文件或文件夹信息

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Java IO与NIO



Java IO中提取文件或文件夹信息，主要通过**File**类来完成。



Java NIO中完成类似的任务，主要通过**Path**类来完成，本讲主要介绍Java IO中操作文件的方法，使用NIO中的相应组件完成同样的任务，将放在NIO部分进行介绍。

File对象的构建



Java IO使用“File”来封装操作系统的文件与文件夹“路径”。



传给File类的构造方法一个路径字符串，就构建了一个File对象。路径字符串中可以使用“/”和“\\”，JDK中会依据当前操作系统自动进行转换：

```
File file = new File("c:\\windows");
```

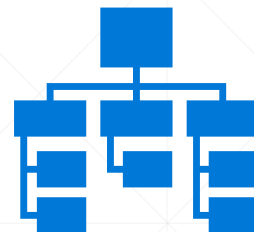
```
File file = new File("c:/windows");
```

} 两者均工作正常



创建了File对象之后，就可以使用它来完成许多与文件和文件夹相关的操作，比如，提取指定文件或文件夹的相关信息。

分解路径字符串



//分解路径字符串

```
public class DecomposeFilePath {  
    public static void main(final String[] args) throws IOException {  
        // "."表示当前路径  
        String path = "./src/DecomposeFilePath.java";  
        File file = new File(path);  
        System.out.println("Absolute path = " + file.getAbsolutePath());  
        // Canonical形式是规范化后的结果  
        System.out.println("Canonical path = " + file.getCanonicalPath());  
        System.out.println("Name = " + file.getName());  
        System.out.println("Parent = " + file.getParent());  
        System.out.println("Path = " + file.getPath());  
        System.out.println("Is absolute = " + file.isAbsolute());  
    }  
}
```

DecomposeFilePath.java

```
DecomposeFilePath x  
"C:\Program Files\Java\jdk-17\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\  
Absolute path = F:\JavaThreadProjects\QueryFileInfo\.\src\DecomposeFilePath.java  
Canonical path = F:\JavaThreadProjects\QueryFileInfo\src\DecomposeFilePath.java  
Name = DecomposeFilePath.java  
Parent = .\src  
Path = .\src\DecomposeFilePath.java  
Is absolute = false
```

获取文件相关的信息

使用File类所定义许多方法，可以查询文件或文件夹的相关信息。

```
GetFileAndDirectoryInfo x
"C:\Program Files\Java\jdk-17\bin\java.exe"
.\src\GetFileAndDirectoryInfo.java 相关信息:
Exists = true
Is directory = false
Is file = true
Is hidden = false
Length = 1172
lastModified = 1663745987590
canExecute =true
canRead =true
canWrite =true

Process finished with exit code 0
```

```
1 usage
private static void printFileInfo(String filePath) {
    File file = new File(filePath);
    System.out.println(file + " 相关信息:");
    System.out.println("Exists = " + file.exists());
    System.out.println("Is directory = " + file.isDirectory());
    System.out.println("Is file = " + file.isFile());
    System.out.println("Is hidden = " + file.isHidden());
    System.out.println("Length = " + file.length());
    System.out.println("lastModified = "+file.lastModified());
    System.out.println("canExecute =" + file.canExecute());
    System.out.println("canRead =" + file.canRead());
    System.out.println("canWrite =" + file.canWrite());
}
```

GetFileAndDirectoryInfo.java